

Benefits of Encapsulation

Some fields $\rightarrow$ read only

Some fields $\rightarrow$ read & write

Why we need private field
public methods get/set

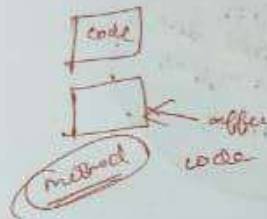
```
Class Emp {  
    private double salary;  
    public double getSalary();  
    public void setSalary();  
}
```

```
    public void setSalary(s)  
    {  
        salary = s;  
    }  
}
```

$\rightarrow$ When there is
change in salary

have
to check only
methods that
mutate it not
Everything

You can chain Interpreter
without objects code



Not to return
Mutable things

which may
given by mutator

It's like giving

key to their

or a complex
machine to child

Accessors

can also make
how to view
data

Metabor
error
checking

Subtype ≤ 0
↓
live errors

some cons

→ present in
different
location

Object.clone() → clone

"clone" so that they
may not clonely
affect it

Class Based privileges

private they can access
by its own class

Private methods

Helper thing
close to impl

Something you
don't want
others to use

only in
method does
not affect
others

Code should not rely
on directory

Private methods

like your routine
should not be
in public

Assured it did
not use elsewhere

Private
things

fields
methods
somebody

methods

not used
elsewhere

Some actions

↓
you should
not done
in public

or not for

public
itself

→ private

Final instance fields

Final → makes it no longer change

instance fields → makes no longer change
once initialized

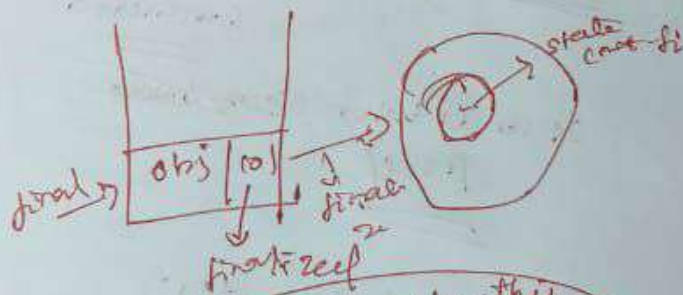
For primitives
immutable

→ It won't change

Mutable
objects

→ does not have
ability to refer to new
object

object it keep can mutate



We cannot change this
reference

Static



You wonder
about global things

Every object has own copy
of instance vars also

But what we have
something common in class

Static

It is shared among class

- Every object shared its copy

It belongs to class

Static fields

Instance
methods
belongs to

It loads when class loads

Itself

private static int nextId

Most of time I recommend
access by class name

Employee - newid:

Static constants

↓
constants we need
among class
like global this

Static constant final static
↓
It cannot be change

Public Static final

↓
They recommend public → Because
Though it accessible we cannot
change since it is final.

Math class

~~public~~

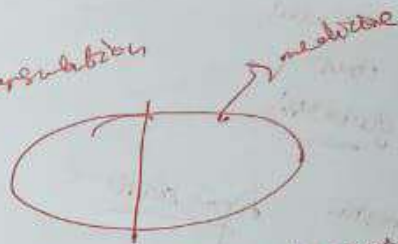
Math.PI

When I need to explain things

1. class → lookie call
definition

2. object → lookie or read
object

3. Encapsulation



hides
impler

↳ You don't know
what method
is

Access → exposes
method

Private methods :- How to cure

How it actually cures

Which this is not required

Private field → what it holds
doesn't matter

~~Public~~

4. Member & Access methods



car

We need to know something

What it has like

brakes
gears
steer } → get it
all

Method

When we start it all
is motor

Check the position
(start)

Private

Have internal
combustion
engine work → start

What is
the engine
diesel?

What
mechanism
work } → start

How engine make (we don't need)

Custom class

You need something
designed
by you

Predefined class

Sometimes you
need already
done by someone

Predefined class

You have to know
what it has what it do?
not how it implements

Custom class

↓
You are going to implement
How it work

fields

→ state

Constructors

→ Initialize when object is
created

methods

↓

do actions

↳ return their

Static methods

Instance fields have

Static methods

Methods for action
done for static fields

Static methods

↓
{ not have explicit
parameter }

Methods

uses only
explicit
parameters

Legal to use objects to call
static methods

Methods
access only
static fields

More recommended

{ Class name }

Static methods

↓
don't have implicit
parameter

(this keyword) X
cannot be used.

main method

```
public static void main (String[] args)
{
}
}
```

Why main method is static?

Because

main method is static

See Static methods don't need objects to call

At when JVM loads there are no objects
if knows classname

↓
main() method

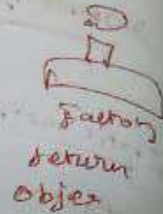
So that's why static main
methods are for the reason!

Factory methods

Factory methods are methods

Construct objects

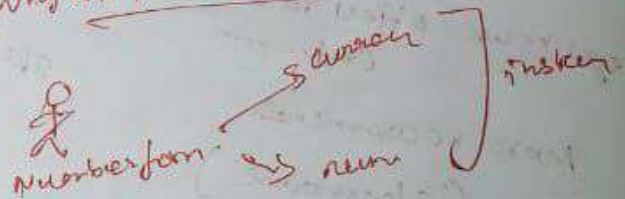
↓
they return object



LocalDate date = LocalDate.now()

↓
factory method

Way use Factory method is



Constructors → same name as class name

so use Factory method

give free definition